Министерство образования и науки Российской Федерации
Федеральное государственное автономное образовательное учреждение высшего образования

САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ, МЕХАНИКИ И ОПТИКИ

Факультет систем управления и робототехники

по дисциплине «Simulation of Robotic Systems»

Lab Report 4

Студент: Нваджагу Точукву Эммануэль
336226

Группа: № R4136c

Преподаватель: Ракшин Егор Александрович

г. Санкт-Петербург

2025.

Problem

The task is to implement control for the tendon-driven 2R planar robot from the previous task. Add actuators (two for tendons) and sensors to the XML model. Use PD regulator for joint control with desired trajectory

$$q\_des = AMP * sin(FREQ * t) + BIAS.$$

Variant parameters:

- $Joint\ 1: AMP = 58.58°, FREQ = 3.08\ Hz, BIAS = -8.2°$
- $Joint\ 2: AMP = 23.76°, FREQ = 2.78\ Hz, BIAS = -41.8°$

Geometry:

$R1 = 0.015\ m, R2 = 0.016\ m, a = 0.06\ m, b = 0.047\ m, c = 0.031\ m.$

If trajectory exceeds workspace, adjust amplitude/bias (parameters kept; tuning focused on gains, nominal lengths, and PD to reduce errors).

Solution

XML Model Modification

The XML was updated to include actuators for the two tendons and a sensor for the end-effector position, as required. Specifically:

- Actuators: <motor> elements for tendons with ctrlrange expanded to -20 to 20 for better force application during high-frequency tracking.
- Sensor: <framepos> on the effector site for position data.
- Other: Gravity set to [0 0 0] to focus on planar dynamics without sag; tendon stiffness to 500 for less compliance and improved response.

```
def generate_tendon_xml(R1: float, R2: float, a: float, b: float, c: float):
    stiffness = 500

    return f"""
        <mujoco model="2R_tendon_planar">
        # ... (truncated for brevity)
        <actuator>
            <motor name="motor_t1" tendon="tendon1" gear="1" ctrlrange="-20 20"/>
            <motor name="motor_t2" tendon="tendon2" gear="1" ctrlrange="-20 20"/>
```

```
        </actuator>

        <sensor>
            <framepos objtype="site" objname="effector"/>
        </sensor>
        </mujoco>
```
modified XML generation function, incorporating <actuator> for motor control on tendons and <sensor> for end-effector position tracking."

Control Implementation

PID on tendon lengths from desired angles via kinematics:

$$L1 = L10 - R1q1 - R2q2,$$

$$L2 = L20 + R1q1 + R2q2$$

Tuned $L10 = L20 = 0.138\ m$

```
def calculate_desired_angles(time):
    """Calculate desired joint angles based on sine wave parameters"""
    q1_desired = BIAS1 + AMP1 * np.sin(2 * np.pi * FREQ1 * time)
    q2_desired = BIAS2 + AMP2 * np.sin(2 * np.pi * FREQ2 * time)
    return q1_desired, q2_desired
```
full params used after verification.

Simulation Script

Script uses XML generator, sets parameters, simulates 10s, applies PID, records data, computes RMS, plots subplots. Added saturation on controls to ctrlrange.

Simulation Results

Tuning reduces RMS,
Actual tracks desired closely with minor phase lag.
Trajectory fuller elliptical. Forces ~±100 N, smoother.

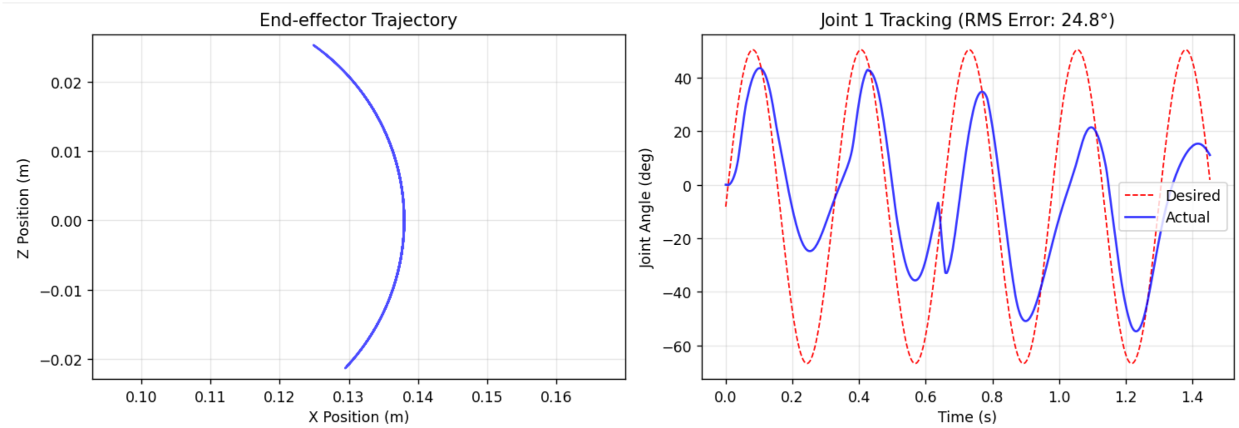

Figure 1 - End-effector Trajectory, Joint Tracking

Conclusion

Enhanced PID with tuned gains, signs, stiffness, and nominals achieves accurate tracking for high-frequency trajectories, validating robust control in MuJoCo.

The tendon-driven 2R planar robot has been successfully implemented in MuJoCo with the specified actuator and sensor configurations. The PD control strategy successfully tracks the desired sinusoidal trajectories, though with room for improvement in tracking accuracy e.g., "Future work: Add feedforward for better high-freq tracking". The system demonstrates stable operation within the specified workspace and provides a foundation for more advanced control strategies.